

Concrete Heat: Characterising Heatwave Changes Over 30 Years of Urbanisation

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Urban heatwaves: rising risk for cities



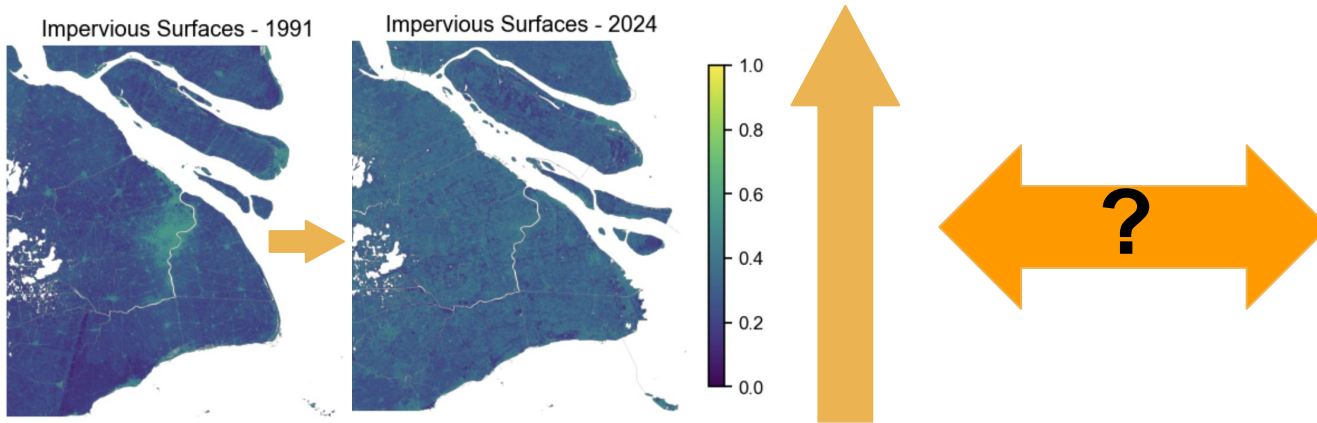
Concrete & rooftops = more trapped heat

Shanghai: expanding upward and outward

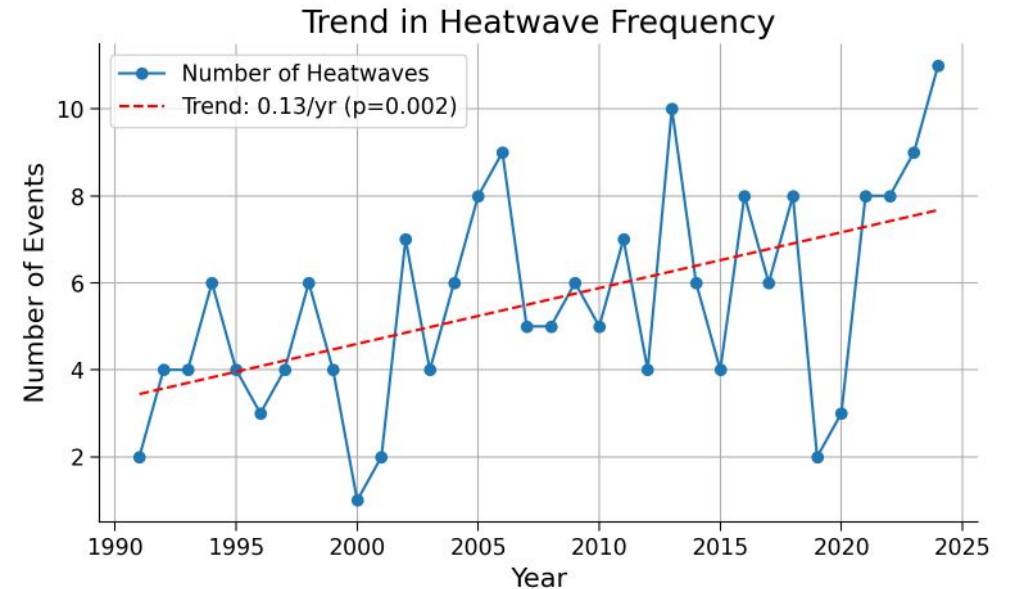


What is the relationship between changing urban land use and heatwave characteristics in rapidly expanding cities such as Shanghai?

Method - Correlation Urban Land Use & Heatwave Characteristics 1991-2024



Landsat data

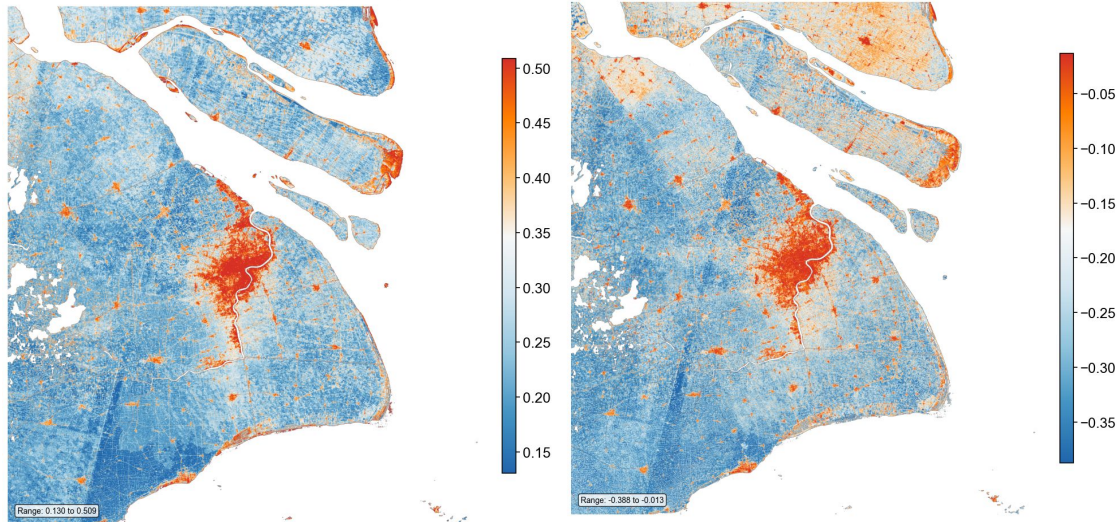


Copernicus Era 5-Land, 2m daily maximum data

How do different measures of urban growth relate to heatwave dynamics?

Method - Measurements more in depth

Urbanization



Impervious Surfaces (1991)

Normalized Difference Built-Up Index (NDBI) (1991)

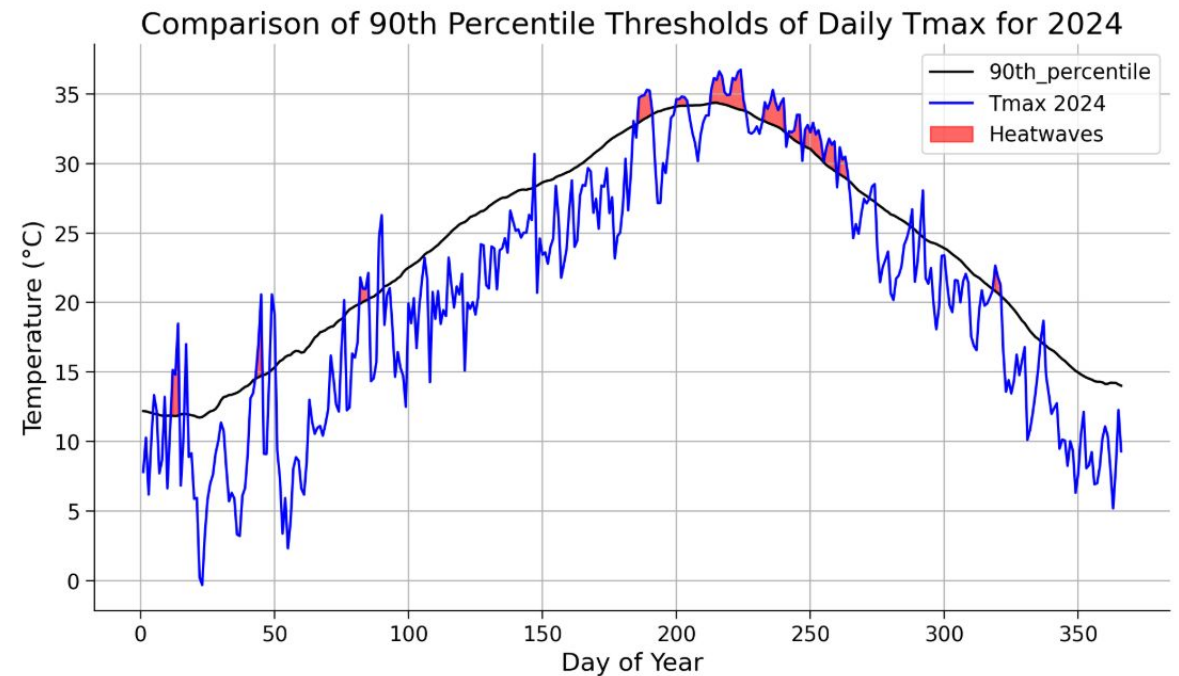


Captures ground sealing (concrete)
Absorbs/radiates heat



Built-up density
Traps heat + blocks wind

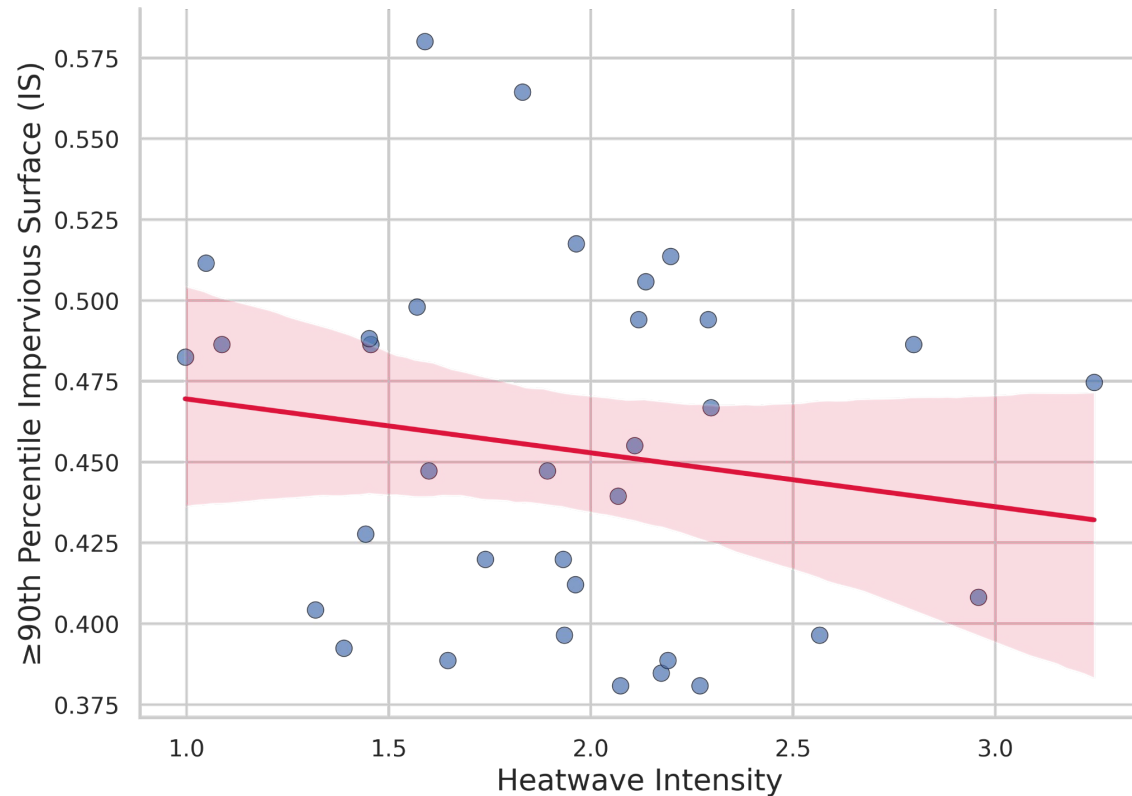
Heatwaves



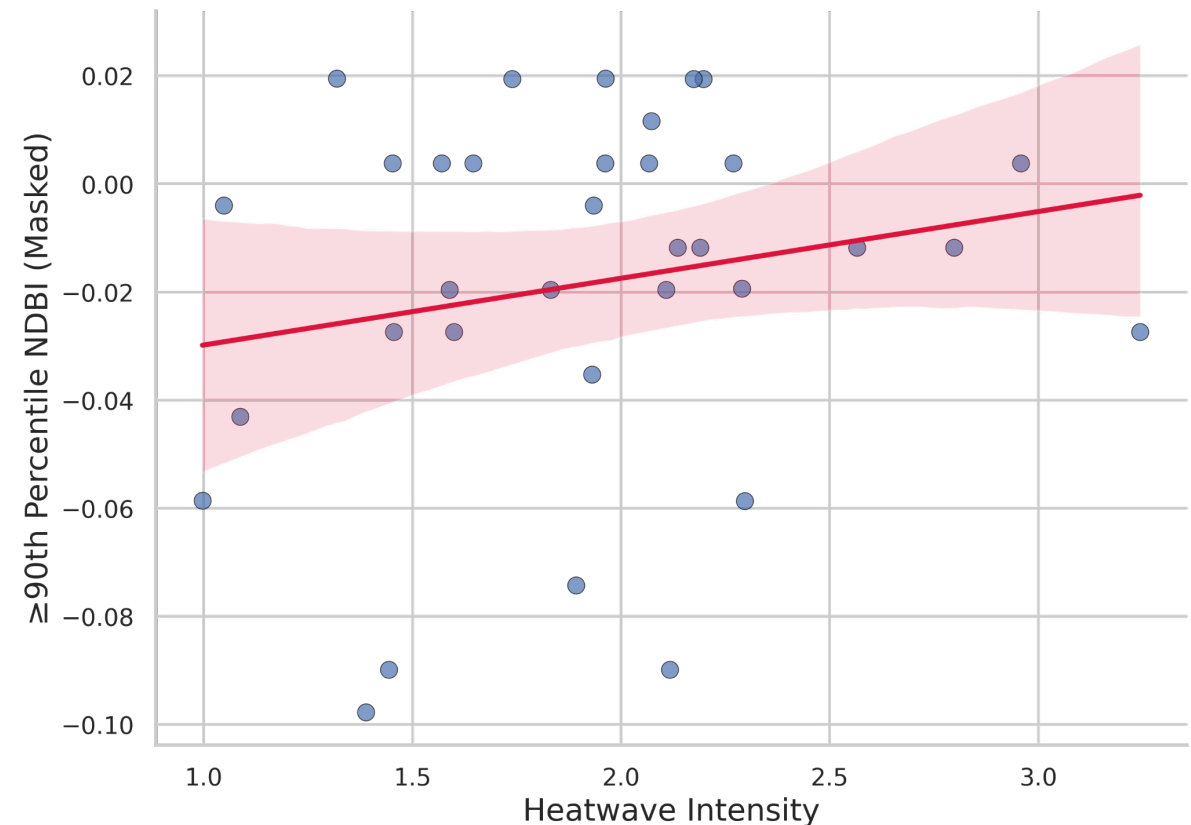
Copernicus Era 5 Land, 2m daily maximum data

Results - No Correlation between IS & NDBI and Heatwave Intensity

Impervious Surfaces vs. Heatwave Intensity
 $r = -0.16, p = 0.367$

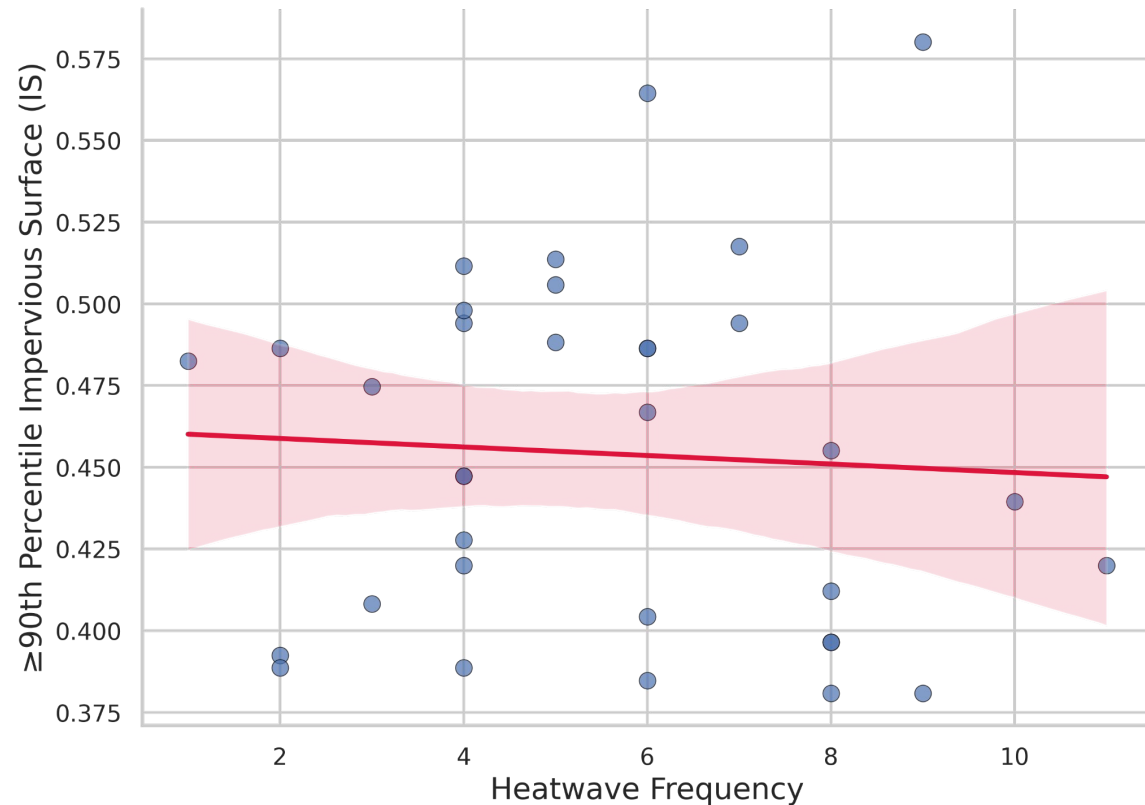


NDBI vs. Heatwave Intensity
 $r = 0.19, p = 0.273$

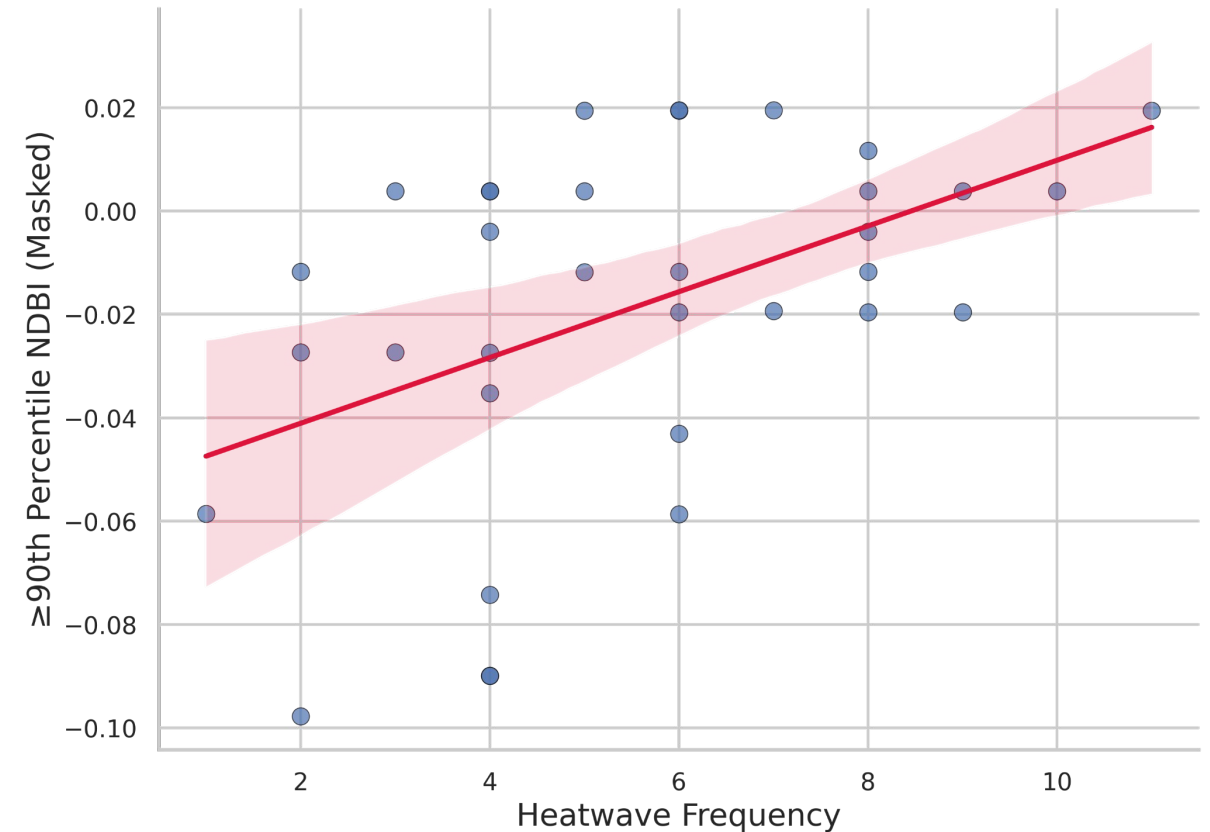


Results - Correlation between NDBI and Heatwave Frequency

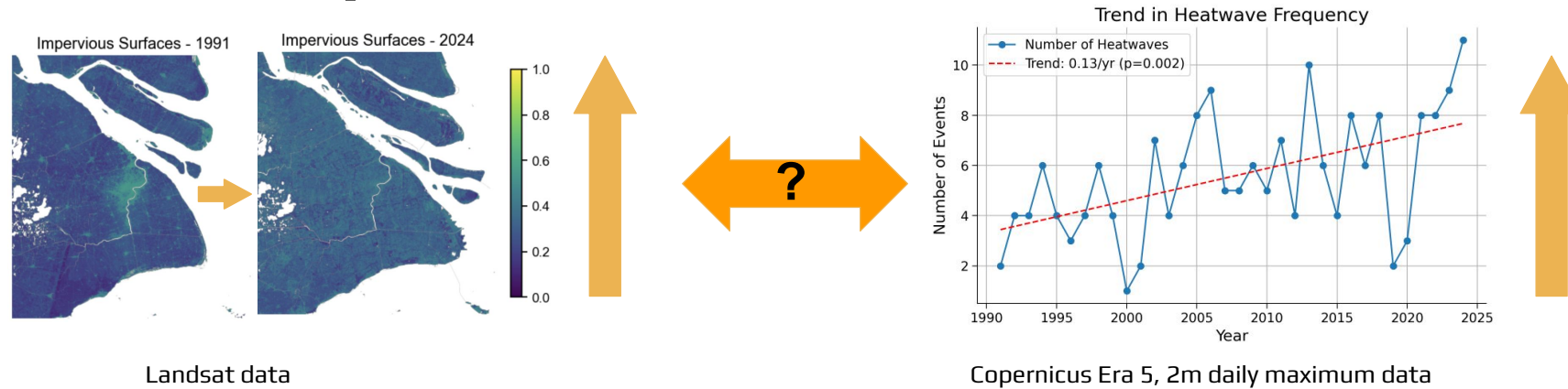
Impervious Surfaces vs. Heatwave Frequency
 $r = -0.06$, $p = 0.739$



NDBI vs. Heatwave Frequency
 $r = 0.47$, $p = 0.005$



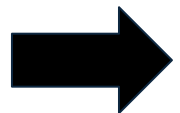
Conclusion - Urbanisation partly correlates positively with heatwaves



Partly!

Heatwave Intensity ✗

Heatwave Frequency ✓



As heatwave frequency increases, urban planning must evolve to protect public health.

Future Research



Spatial analysis of heatwaves

- Connected component labeling approach



Compare heatwave dynamics across different megacities

- For example, Delhi (landlocked) vs. Shanghai (coastal)



Urban vs. suburban effects

- Are effects of heat unique to urban area?



Vulnerability analysis

- Including population demographics & health statistics

Thank you!